

WM-D6C Servo Replacement Module — DSR-1

Revision 0.2 | Hardware + Firmware Module Data Sheet
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Document Number	DSR-1
Revision	0.2 — Battery-integrated configuration added; LiPo/BQ24074/DW01/two-board architecture
Status	Design Phase — Pending Bench Verification
Target Machine	Sony Walkman WM-D6C (primary testbed: serial SN72795, main board suffix -12)
Primary MCU	STM32G0C1KCU6 — UFQFPN32, 64 MHz Cortex-M0+, 256 KB flash, GP variant
Repository	github.com/ChipCreates/wmd6c-servo
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Firmware License	MIT License
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1. Introduction and Background

1.1 The Sony WM-D6C

The Sony Walkman WM-D6C is a professional-grade portable cassette recorder produced by Sony from approximately 1984 through the early 1990s. Unlike consumer Walkman products, the WM-D6C was designed for broadcast journalists, field recordists, and professional audio applications, featuring full-size record and playback heads, Dolby B and C noise reduction, a precision capstan servo system, and a wow-and-flutter specification of 0.05% WRMS — exceptional for a handheld battery-powered device of its era.

The machine achieved its speed stability through a dedicated capstan servo IC, the Sony CX20084, which formed the heart of a phase-locked loop comparing the frequency-generator (FG) pulse train from the capstan motor against a 34.7 kHz crystal reference divided down by the MSM58141RS counter IC.

1.2 The Failure Mode This Module Addresses

The WM-D6C's DC power input connector (CN301) is a non-standard 5.5mm barrel jack wired with negative polarity on the centre pin — the opposite of virtually every generic DC adapter sold since the 1980s. When a standard centre-positive adapter is connected, the full adapter voltage is applied in reverse across the B+1 supply rail, which feeds the CX20084 servo IC directly with no protection. The IC fails instantly. The CX20084 is a proprietary Sony ASIC out of production for decades; replacements are scarce and of uncertain provenance.

⚠ WARNING The original CN301 power jack is the single most common cause of electronic failure in surviving WM-D6C units. This module eliminates the hazard permanently by replacing CN301 with correctly protected power input in every variant.

1.3 Module Purpose and Scope

This module simultaneously replaces three components:

- The CX20084 capstan servo IC (IC601) — replaced by digital servo firmware on the STM32G0C1KCU6

- The CP304 DC-DC boost converter (6V → 10.8V motor rail) — replaced by an MT3608-based boost on the same PCB
- The CN301 DC power input jack — replaced by USB-C PD (Variant A), a polarity-agnostic barrel interface (Variant B), or eliminated in the battery-integrated build where the battery daughter board is the power source

All other WM-D6C circuitry is retained: Q601, Q703, Q704, FG901, RV601/602/603, S601, and the Auto-Off board. The module interfaces through a keyed harness to the original CP304 connector and the CX20084 solder pads.

1.4 Design Philosophy

Cost and simplicity of installation drive adoption. A module costing more than approximately \$35 landed, or requiring specialised skills, will not spread. Every component choice is evaluated against this constraint.

Open source hardware, firmware, and documentation ensure the community can maintain, adapt, and extend the design indefinitely. All files are released under maximally permissive open licences (CERN OHL-P v2, MIT, CC BY 4.0).

Technical correctness without compromise means the digital servo loop must equal or exceed the original CX20084 in speed stability, and the power supply must be more robust in every measurable respect.

2. Functional Overview

2.1 System Architecture

For the two wall-powered variants (A and B), the module is a single two-layer PCB occupying the CP304 cavity, dividing functionally into four zones: the variant-specific power input zone; the power management zone (MT3608, MCP1700, TPS63070 — see Section 5); the STM32 microcontroller zone; and the machine interface zone (J1, an 8-position 1.25mm JST connector).

The battery-integrated configuration — the primary, self-contained, portable build — is a two-board design. A dedicated power daughter board mounted behind the battery bay carries the entire power subsystem (USB-C input, BQ24074 power-path charger, LiPo pack connections, DW01 + dual-FET protection, and the TPS63070 boost). The daughter board injects regulated 6.0V at the original battery-terminal node on the main board, ahead of S901, so the front-panel power switch remains the machine's real on/off control. In this build the servo module's power-input zone is left unpopulated; the module receives B+1 via the harness. Section 5.4 gives full detail.

2.2 Variant Differentiation

The two wall variants share an identical PCB gerber, BOM structure, and STM32 firmware, differing only in which components are populated in the power input zone.

Parameter	Variant A — USB-C PD	Variant B — Barrel Jack
Power Input	USB-C Type C receptacle, panel mount or flex tail to CN301 opening	5.5mm barrel jack, CN301 retained in original panel opening
Input Voltage	9V negotiated via USB PD (IP2721 trigger IC)	4.5V to 9V, either polarity accepted
PD Negotiation	IP2721 hardware PD trigger — no firmware required	Not applicable
Polarity Protection	Inherent in USB-C physical format	LTC4359 ideal diode bridge controller, four-MOSFET topology
Overcurrent	USB PD source current limiting	500mA polyfuse (MF-R050 or equivalent, resettable)
Overvoltage	PD contract voltage capped at 9V by IP2721	SMBJ7.0A TVS diode, 7V standoff, transient clamp

Input bulk storage	220µF / 10V electrolytic (motor surge holdover)	220µF / 10V electrolytic (bridge transition holdover)
CN301 Disposition	Removed — opening modified for USB-C	Retained in original panel position

NOTE Beyond the two wall variants, the battery-integrated configuration is the primary self-contained build — documented in Section 5.4. It provides ~3–4 h cordless runtime and charges in place over USB-C.

2.3 Power Rails

Rail	Consumers	Regulation	Current
B+1 (6.0V)	Audio circuits, logic, servo reference, Auto-Off board	5.85V – 6.15V	600mA continuous
B+3 (10.8V)	Capstan motor M901 via Q703/Q704 switching	10.5V – 11.0V	500mA peak (acceleration)
VDD (3.3V)	STM32G0C1KCU6, USB PHY (VDDIO2 tied internally in GP variant)	3.27V – 3.33V	250mA

3. Microcontroller — STM32G0C1KCU6

3.1 Device Selection Rationale

The STM32G0C1KCU6 is selected as the central processing element after evaluation against the functional requirements. It is a 32-bit Arm Cortex-M0+ microcontroller in the STM32G0 family, operating at up to 64 MHz with 256 KB of flash and 144 KB of SRAM. The GP variant (KCU6) has VDDIO2 tied to VDD internally, eliminating the need for a separate VDDIO2 supply in a single-3.3V system.

Key selection factors: crystal-less USB 2.0 Full Speed (HSI48 + SOF trimming — no external crystal or bridge chip required); built-in UCPD1 peripheral on PA8/PA9 for PD contract monitoring without external hardware; TIM3 PWM for filtered motor drive; 12-bit ADC with three simultaneous channels for the speed-trim pots; and TIM2 32-bit input capture for the FG signal at 64 MHz resolution (~15.6 ns per tick). The UFQFPN32 5×5mm QFN package is within standard JLCPCB SMT capability.

The 256 KB flash (superseding the earlier 128 KB KBU6 selection) provides headroom for the servo firmware, USB CDC stack, DFU bootloader, configuration storage, and future enhancements including the LiPo fuel-gauge OCV table and the battery-build LED driver.

3.2 Key Specifications

Parameter	Value
Full Part Number	STM32G0C1KCU6
Package	UFQFPN32 — 5.0 × 5.0 mm, 0.5 mm pitch QFN, exposed VSS pad
Core	Arm Cortex-M0+, 32-bit, single-cycle multiply
Core Clock	64 MHz maximum (HSI16 × PLL)
Flash Memory	256 KB with read-while-write capability
SRAM	144 KB (includes 32 KB USB PMA)

ADC	12-bit, up to 12 channels, simultaneous sample
DAC	12-bit, 2 channels (1 used for motor drive)
Timers	TIM2 (32-bit, input capture, FG), TIM3 (16-bit, PWM Option B)
USB	Full Speed 2.0 device, crystal-less operation via HSI48 + SOF trim
UCPD	Built-in USB Type-C PD controller — CC1/CC2 on PA8/PA9
VDDIO2	Tied to VDD internally in GP variant — no external supply required
Supply	1.7V – 3.6V (operated at 3.3V)

3.3 Clock Configuration

The STM32 uses its internal HSI16 (16 MHz RC oscillator) multiplied through the PLL to reach 64 MHz for the CPU and peripherals. For USB, the internal HSI48 (48 MHz RC) is enabled and automatically trimmed against the USB Start-of-Frame signal by the CRS (clock recovery system) peripheral, achieving the ± 2500 ppm accuracy required for USB Full Speed without any external crystal. No external oscillator components are required anywhere on the board.

3.4 Pin Assignment

Pin / Peripheral	Signal	Function
PA0 / TIM2_CH1	FG_IN	FG pulse capture — ISR computes period; digital PLL input
PA1 / ADC_IN1	ADC_RV601	Base speed trim RV601 wiper → target period offset
PA2 / ADC_IN2	ADC_RV602	Speed-Tune slider RV602 wiper → user $\pm$ speed offset
PA3 / ADC_IN3	ADC_RV603	Speed range trim RV603 → scales RV602 range
PA4 / DAC1_OUT1	MOTOR_PWM	Motor drive PWM output (PA6 TIM3 + RC + NPN)
PA5 / GPIO	MOTOR_EN	Motor enable input from IC601 pin 7 (via divider)
PA6 / TIM3_CH1	PWM_OUT	PWM motor drive output (Option B with NPN level-shift)
PA7 / GPIO	S601_GPIO	Speed-Tune switch S601 position
PA8 / UCPD1_CC1	CC1	USB-C CC1 — PD contract monitoring (Variant A)
PA9 / UCPD1_CC2	CC2	USB-C CC2 — PD contract monitoring (Variant A)
PA9 / USART1_TX	UART_TX	Debug UART test pad (optional)
PA11 / USB_DM	USB_DM	USB D- (USB CDC / DFU)
PA12 / USB_DP	USB_DP	USB D+ (USB CDC / DFU)
PA13 / SWDIO	SWDIO	SWD debug/programming
PA14 / SWDCLK	SWDCLK	SWD debug/programming clock
ADC (battery build)	ADC_VBAT	VBAT_SENSE from power daughter board — LiPo fuel gauge
GPIO (battery build)	CHG_STAT	BQ24074 CHG/PGOOD — charge animation control
GPIO (battery build)	S801_SENSE	S801 BATT-position mode sense — enables LED fuel gauge drive

GPIO/PWM ×5 (batt)	D801–D805	LED bar 1–5 segment LiPo SoC fuel gauge (BATT mode)
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4. Capstan Servo Loop

4.1 Theory of Operation

The digital PLL servo replaces the CX20084 analogue phase-locked loop with a software equivalent running entirely within the STM32. TIM2 captures the rising edges of the FG901 optical sensor output, providing a precise measurement of each FG pulse period in units of the 64 MHz timer clock (~15.6 ns per tick). The firmware computes the error between the measured period and the target period, accumulates the error through a PI controller, and drives the motor base transistor Q601 via the DAC or PWM output to close the loop.

The target FG period is derived from the known tape speed (4.75 cm/s) and the FG disc geometry; the exact FG_TARGET_HZ is measured on the physical unit with a test tape. The servo loop runs at the FG pulse rate (~2500 Hz at normal speed), which is well above the mechanical bandwidth of the transport but below any structural resonance.

4.2 FG Signal Conditioning

FG901 (GP2S22AB) is an open-collector optical interrupter; a pull-up resistor on the WM-D6C main board produces a voltage swing at J1 pin 1. A resistor divider (R3/R4 — values pending bench measurement of the actual swing) scales the signal to a level safe for the STM32's 3.3V input. BAT54 Schottky diodes clamp the node to VDD and GND for overvoltage and reverse protection. A small C0G capacitor at the ADC node provides HF filtering.

4.3 PI Control Algorithm

The servo loop uses a fixed-point Q16 PI controller (bare-metal C; no float, no HAL, no RTOS, no malloc). PNP convention: a longer measured period means the motor is running too slowly, which requires more base drive (lower base voltage on Q601). The PI output clamps to the safe PWM range.

```
// Q16 fixed-point PI — PNP convention
int32_t error = measured_period - target_period;
// Positive error = too slow = need more drive = lower base voltage
integral += error;
integral = clamp(integral, -INTEGRAL_LIMIT, INTEGRAL_LIMIT);
int32_t output = CENTER - KP_Q16 * error - KI_Q16 * integral;
output = clamp(output, OUT_MIN, OUT_MAX);
```

Starting constants KP and KI are tuned live via USB CDC on the bench. Settings are saved to flash.

4.4 Motor Drive Output Stage

Rejected PA4 DAC Drive

Used when Q601 base voltage $\leq 3.3\text{V}$ (confirm on bench). PA4 PWM output → series resistor → Q601 base. A pull-up to +3V3 holds Q601 off at boot before firmware starts.

Option B — PWM + RC + NPN Level Shift

Used when Q601 base voltage $> 3.3\text{V}$. PA6 PWM → RC filter → NPN (MMBT3904) → Q601 base. Only one option is populated; the other is DNP.

4.5 Speed Reference Potentiometers

Three pots connect to ADC inputs via J1 pins 4/5/6. RV601 (47kΩ cermet) provides a fine base-speed trim offset. RV602 (20kΩ carbon slider) is the user-accessible Speed Tune control. RV603 (47kΩ cermet) scales the range of RV602. All three wiper voltages must be verified on the bench ($\leq 3.3\text{V}$) before connecting to the ADC; a divider is added if any exceeds this.

5. Power Supply Architecture

5.1 Shared Power Management

5.1.1 MT3608 Boost Converter — 6V to 10.8V (B+3 Motor Rail)

The MT3608 synchronous step-up regulator in SOT-23-6 replaces the CP304 potted boost module. It accepts B+1 (6.0V) and produces the 10.8V B+3 rail for motor M901 via Q703/Q704. Output is set by a resistor divider on the feedback pin. A 4.7 μ H inductor and SS14 Schottky rectifier complete the boost topology. Switching frequency ~1.2 MHz keeps noise far above the audio band; output ripple at 10.8V with 22 μ F is under 50mV peak-to-peak.

5.1.2 MCP1700 LDO — 6V to 3.3V (MCU Supply)

The MCP1700T-3302E/TT (SOT-23-3, 250mA, 178mV dropout at 100mA) produces 3.3V for the STM32. Regulation holds to ~3.5V input — well below the 4.5V where the WM-D6C machine itself operates marginally. Input and output are decoupled per the datasheet recommendation.

5.2 Variant A — USB-C Power Delivery Input

5.2.1 IP2721 USB PD Trigger

The IP2721 (SOT-23-6) negotiates a fixed 9V PDO from a USB PD source without requiring firmware — a single CFG resistor selects the voltage. If the source cannot supply 9V, the IP2721 falls back to 5V, which is insufficient to operate the machine, preventing under-voltage damage. Once 9V is established, it drives a TPS63070 buck regulator (Section 5.1.2 of the shared core now re-used as a buck, not boost).

5.2.2 TPS63070 Buck-Boost — 9V to 6V (Variant A)

In Variant A the TPS63070 operates in buck mode, converting 9V PD input to the regulated 6.0V B+1. The same part serves as a boost converter in the battery-integrated build (Section 5.4). The output voltage is set by a resistor divider ($R_{upper} = 649k\Omega$, $R_{lower} = 100k\Omega$, 0.8V reference $\rightarrow$ 6.0V output), common to all configurations.

5.2.3 USB Data Path

The USB-C connector carries both the PD CC lines (PA8/PA9 $\rightarrow$ UCPD1) and the USB 2.0 data lines (PA12/PA11 through a USBLC6-2SC6Y ESD array). The STM32 UCPD1 monitors the PD contract; D+/D- carry the USB CDC live-tuning and DFU channels.

5.3 Variant B — Barrel Jack Input

5.3.1 Polarity-Agnostic Input Stage

The LTC4359 ideal diode bridge controller drives four P-channel MOSFETs in a full-wave bridge configuration. Regardless of the polarity at CN301, the bridge output is always correct polarity with a < 50mV drop at rated current, versus ~1.2V across a conventional diode bridge — unacceptable given the 6V nominal input and 5.85V minimum machine requirement.

5.3.2 Protection Stack

- Resettable polyfuse (500mA hold, 1.5A trip) — limits current during fault; auto-resets when cleared
- SMBJ7.0A TVS diode (7V standoff) — clamps motor back-EMF transients and ESD events
- 220 μ F / 10V electrolytic input bulk capacitor — reservoir for motor-startup surge and bridge-transition holdover
- 100nF ceramic input bypass — HF decoupling at power entry

5.4 Battery-Integrated Configuration — Two-Board Design

5.4.1 The LiPo Pack — 3 Cells in Parallel (3P)

The runtime source is three LiPo cells wired in parallel. The reference cell is the 803040 form factor (8 $\times$ 30 $\times$ 40 mm, 3.7V, 1000mAh, on-cell PCM, PH2.0 pigtail), giving a pack that is 3.7V nominal, ~4.2V fully charged, ~3.0V usable floor, 3000mAh / ~11.1Wh total. Three cells stack flat at 40 $\times$ 30 $\times$ 24mm within the ~57 $\times$ 32 $\times$ 29mm battery cavity, leaving ~4.6mm clearance on the stacking axis. That clearance is the swell budget — LiPo cells expand perpendicular to the flat faces, and it must be left empty. Cells must be matched to within ~30–50mV of resting voltage before first parallel join; the three on-cell PCMs do not prevent inter-cell inrush at the moment of

connection. The cells plug into the daughter board, making the pack removable for service. (Supersedes 1S4P IFR14500 LFP; LiPo chosen for off-shelf availability in required form factor.)

Parameter	Value / Description
Configuration	Two-board: power daughter board (behind battery bay) + servo module (CP304 cavity)
Cells	3× 803040 LiPo in 3P — 3.7V, 3000mAh, 11.1Wh; cells plug into daughter board (service disconnect)
Cavity dimensions	~57 × 32 × 29 mm (≈52 cm³); cells stack 40×30×24mm; ~4.6mm swell clearance (must remain empty)
Charger IC	TI BQ24074 — power-path with DPPM; supplement mode for motor surge
Charge current	R_ISET ≈ 1.1 kΩ → ~0.8A (~0.27C)
Input current limit	R_ILIM ≈ 800 Ω → ~2A; DPPM auto-balances deck vs charge
USB input	Plain 5V VBUS — no PD (9V would over-volt charger)
Pack protection	DW01 + FS8205 dual-FET pack-level protection (+ on-cell PCMs as first layer)
B+1 regulator	TPS63070 buck-boost on daughter board; SYS → 6.0V boost mode (D ≈ 0.38)
B+1 injection point	Original battery-terminal node on main board (ahead of S901 power switch)
S901 switch	Preserved as native machine power switch (injection on battery side)
Module role	Receives B+1 (J1 pin 7 = input); power-input zone unpopulated; servo + MT3608 + MCP1700 unchanged
Board-to-board link	B+1, GND, VBAT_SENSE → STM32 ADC, BQ24074 CHG/PGOOD → STM32 GPIO, (opt) USB D+/D-
Fuel gauge	STM32 reads VBAT_SENSE; LiPo OCV table; drives D801–D805 as 1–5 segment bar (S801=BATT)
Runtime	~3–4 h per charge at ~2.5–3W average draw; charge-in-place while machine plays
Supersedes	Earlier 1S4P IFR14500 LFP pack with CN3058E + HY2112-CB (LiPo chosen for off-shelf availability)

5.4.2 BQ24074 Power-Path Charger

The BQ24074 (TI, single-cell Li-ion/LiPo linear charger with integrated Dynamic Power Path Management) handles USB-present operation correctly: it powers the SYS load from USB first and routes only surplus current to charging (DPPM); lets the battery assist the SYS load during motor-startup surge that briefly exceeds the input current limit (supplement mode); and hands SYS off to the battery seamlessly when USB is removed. A programmable input current limit (ILIM) protects a weak USB source, and thermal regulation folds back charge current if the IC heats up — relevant on the small daughter board. This replaces the earlier CN3058E + discrete load-share P-FET approach, which had no power-path capability.

Charge current: R_ISET ≈ 1.1kΩ → ~0.8A (~0.27C into 3000mAh, gentle and low-dissipation; P_diss ≈ 0.8W at 5V in). Input limit: R_ILIM ≈ 800Ω → ~2A, matching a 5V/2.4A-or-better source; DPPM auto-balances deck vs charge without hand-partitioning. The TS pin is defeated with a fixed resistor pair (cells have no external thermistor); CHG and PGOOD status outputs are routed to the STM32 GPIO for charge animation on the LED fuel gauge.

5.4.3 Pack Protection — DW01 + FS8205

A DW01 protection IC drives a dual N-channel FET (FS8205-class) in series between the pack and the daughter board. For LiPo, the DW01's thresholds (over-charge 4.25V, over-discharge ~2.4V) are correct. Note: an earlier LFP revision of this design used an HY2112-CB and explicitly warned against the DW01 because its thresholds were wrong for the 3.6V LFP chemistry. With the change to LiPo, the DW01 is now the right part.

5.4.4 TPS63070 Boost Regulator — SYS to 6.0V (Battery Build)

On the daughter board, the same TPS63070 buck-boost (same divider, same layout) runs in boost mode, converting the BQ24074 SYS rail (~3.0–4.9V) to a regulated 6.0V. At the nominal 3.7V pack voltage, $D \approx 0.38$; at the 3.0V floor, $D \approx 0.50$. Peak inductor current at worst-case ~4W load, 3.0V input, 90% efficiency is ~1.5A — under the TPS63070's 3.6A switch limit.

5.4.5 6V Injection at Battery Terminals and S901

The daughter board's regulated 6.0V output is wired to the original battery-positive terminal location on the main board (spring contacts removed during installation). Because the stock wiring runs battery → S901 → B+1 net, this injection point is on the battery side of S901, so the front-panel power switch remains the machine's real on/off control. A series 1N5819-class Schottky at the injection point provides reverse protection; its ~0.3V drop puts ~5.7V on B+1, within operating specification.

5.4.6 Board-to-Board Interface

The following conductors link the daughter board to the servo module, separate from the J1 harness: B+1 and GND (the regulated 6V power path, routed via the battery terminals and the machine's own B+1 net to the harness); VBAT_SENSE (pack voltage, for the STM32 fuel gauge — 22kΩ/100kΩ divider, see note on divider sizing for 4.2V LiPo peak); CHG/PGOOD (BQ24074 status, for charge animation); and optionally USB D+/D- (if CDC tuning is routed from the daughter board USB-C rather than a separate bench header on the module).

5.4.7 Battery Level Indicator

With B+1 held at a regulated 6.0V, the stock BATT-position single-LED brightness dimming (Q801 off the B+1 divider) no longer reflects the pack charge. The DSR-1 re-references it: the STM32 reads VBAT through VBAT_SENSE, estimates state of charge from a LiPo open-circuit-voltage lookup table (the sloped LiPo discharge curve, 4.2V → 3.0V, makes voltage-based SoC reliable), and drives the five-LED bar D801–D805 directly as a 1–5 segment fuel gauge while S801 is in BATT. The STM32 uses the BQ24074 CHG/PGOOD status to animate segments upward while charging, hold full on charge completion, and flash the bottom segment below ~15%.

6. USB Interface and Firmware

6.1 USB CDC Virtual COM Port

The STM32 presents a USB CDC (Communications Device Class) virtual serial port on D+/D-. The USB clock is derived from the internal HSI48 oscillator with CRS automatic trimming from the USB Start-of-Frame signal — no external crystal is required. The device presents as a standard CDC ACM device on all platforms.

6.2 Live Tuning Command Interface

While the servo is running, the following single-character commands are accepted over USB CDC:

- p+ / p- — adjust KP ±0.01
- i+ / i- — adjust KI ±0.001
- t — print telemetry snapshot (FG period, target, error, integral, DAC value, three ADC readings)
- s — save current constants to flash
- r — reset constants to compiled defaults

6.3 Firmware Update via USB DFU

The STM32G0C1KCU6 supports USB DFU (Device Firmware Update) in ROM bootloader, entered by asserting BOOT0 high at reset. No ST-Link or SWD connection is required for firmware updates in the field. The SWD header (PA13/PA14) is retained for initial programming and debugging during development.

6.4 Firmware Architecture

The firmware is plain bare-metal C compiled with arm-none-eabi-gcc. There is no HAL, no LL, no RTOS, no malloc, and no C++ anywhere in the codebase. Every allocation is static. The implementation constraints are intentional: determinism, auditability, and community maintainability are all improved by avoiding abstraction layers.

- Clock init: HSI16 × PLL → 64 MHz; HSI48 + CRS for USB
- TIM2: 32-bit free-running at 64 MHz; CH1 input capture on PA0 rising edge; ISR computes FG period
- PI controller: Q16 fixed-point arithmetic; clamped integral; sub-500ns ISR execution
- DAC / PWM: initialised at mid-scale; Q601 held off until servo loop starts
- ADC: three channels scanned in sequence; results low-pass filtered
- USB CDC: virtual COM; command parser in main loop
- Flash storage: PI constants and target period saved to last flash page

7. WM-D6C Machine Interface

7.1 Interface Connector J1

J1 is an 8-position 1.25mm pitch JST connector on the module. A pre-assembled harness with keyed housing connects to the original CP304 harness socket on the WM-D6C main board and to the supplementary wires soldered to the removed CX20084 pads.

Pin	Net	Dir	WM-D6C Connection
1	FG_RAW	Machine → Module	FG901 optical sensor output
2	Q601_BASE	Module → Machine	CX20084 pin 15 pad (motor drive output)
3	MOTOR_EN	Machine → Module	IC601 pin 7 net (Auto-Off board enable)
4	RV601_WIPER	Machine → Module	Base speed trim RV601 (47kΩ cermet)
5	RV602_WIPER	Machine → Module	Speed-Tune slider RV602 (20kΩ carbon)
6	RV603_WIPER	Machine → Module	Speed range trim RV603 (47kΩ cermet)
7	B+1 / VBATT	↔ (see note)	6V B+1 rail — module drives (wall variants) / receives (battery build)
8	GND	Shared	WM-D6C chassis ground

NOTE J1 pin 7 direction: in wall variants (A and B) the module drives B+1 out through this pin. In the battery-integrated build the daughter board supplies B+1 via the battery terminals and the machine's own B+1 net; the module receives B+1 through this pin (power-input zone unpopulated).

7.2 Signal Voltage Compatibility

All WM-D6C signals on J1 pins 1–6 are sourced from circuits referenced to the 6V B+1 rail. The STM32 operates at 3.3V. Signals requiring level adaptation:

- FG_RAW (pin 1): R3/R4 voltage divider scales the FG swing to ≤ 3.3V; BAT54 Schottky clamps protect against overshoot. Divider values depend on measured swing — verify on bench before populating.
- MOTOR_EN (pin 3): IC601 pin 7 net is held at ~4.4V during playback. A 10kΩ/22kΩ divider reduces this to ~3.03V, within the STM32 input range.

- RV601/602/603 (pins 4–6): pot wipers are referenced to the B+1 rail. Verify measured wiper voltage range on bench; add dividers if any exceeds 3.3V.

7.3 Components Removed from WM-D6C

Ref	Part	Reason
IC601	CX20084	Replaced by STM32 digital servo
X701	34.7 kHz crystal	Servo reference no longer needed
IC701	MSM58141RS	Crystal divider/comparator no longer needed
CP304	Boost module (potted)	Replaced by MT3608 on module
CN301	Barrel jack (negative centre)	Opening repurposed for power input per variant
Battery spring contacts	AA cartridge terminal springs	Removed in battery build (corroded in many units); daughter board feeds 6V here

7.4 Components Retained and Reused

- Q601 (package/marketing pending): motor-control device — bench-verify package, polarity, and base range before install
- Q703 / Q704: motor switching transistors — untouched
- FG901 (GP2S22AB): optical speed sensor — stays in circuit
- RV601 / RV602 / RV603: speed trim pots — stay wired to J1 pins 4/5/6
- S601: Speed-Tune switch — stays on PA7
- D801–D805 / IC801 (CX10043) / S801: LED bar — re-used by STM32 as LiPo fuel gauge in battery build

8. PCB Design

8.1 Board Specification

Parameter	Value
Target size (servo module)	20 × 25 mm (fits CP304 cavity)
Layers	2-layer
Finish	ENIG (gold)
Copper weight	1 oz per layer
Board thickness	1.6 mm FR4
Minimum trace/space	0.1 mm / 0.1 mm
Minimum via drill	0.3 mm
Fabricator	JLCPCB (2-layer, standard process)

8.2 Layout Guidelines

- Place the MT3608 inductor and SS14 diode as close together as possible, with short high-current traces
- Keep the TPS63070 inductor directly adjacent to its LX1/LX2 switch pins
- Decouple STM32 VDD pins with individual 100nF 0402 ceramics placed within 0.5mm of each pad

- Route the FG_IN trace away from switching power traces and motor drive signals
- Place the PWM output RC filter (if present) close to Q601 connection point
- Ground pour on both layers; stitch with vias every ~3mm on the primary GND pour
- Keep USB D+/D- trace length matched to within 0.5mm, with 90Ω differential impedance

9. Installation Procedure

The following procedure applies to a Sony WM-D6C that has already received full mechanical service (belt, pinch roller, FG disc cleaning, capstan bearing lubrication). Mechanical service precedes electronic diagnosis.

⚠ WARNING Before opening the machine, verify the original fault using a calibrated test tape (motor runs fast / uncontrolled). Complete mechanical service and retest. Proceed with module installation only if the fault persists.

9.1 Tools Required

- Phillips #00 and #000 screwdrivers
- Soldering iron, fine tip, temperature-controlled, set to 320°C
- Rosin-core solder 0.5mm; solder wick fine gauge
- IPA (≥90%) and cotton swabs; white vinegar for alkaline corrosion neutralisation
- Multimeter (continuity and DC voltage)
- Fine-point tweezers; small flat-blade screwdriver for harness connectors

9.2 Pre-Installation Checks

- Test Q601 (package pending) in diode mode: B-E and B-C junctions should read ~0.6–0.7V. Zero or open = replace.
- Measure the B+3 rail at CP304 output on battery power. ~10.8V = CP304 still functional; zero or very low = module MT3608 will take over, no further action.
- Measure the FG901 signal swing at IC601 pin 13 with a multimeter on AC. A reading above zero confirms FG circuit is operational.

9.3 Installation Steps

- Open the WM-D6C. Remove the four case screws (two each side). Separate upper and lower panels.
- Unplug the CP304 harness connector from the main board. Remove the CP304 module from its cavity by releasing the retaining claw.
- Locate IC601 (CX20084) on the main board. Using solder wick, carefully remove IC601. Clean pads with IPA. Identify pads 13 (FG), 14 (GND), 15 (motor drive), 16 (VDD).
- Locate X701 crystal and IC701 on the Auto-Off board. Desolder and remove both.
- For Variant A only: desolder CN301 from the lower panel. Modify the panel opening for the USB-C connector.
- Connect the module harness to J1. Route supplementary harness wires (FG, motor drive, GND) to the IC601 pads. Plug the main harness into the original CP304 connector socket.
- For Variant B: connect barrel jack wires from CN301 to the Variant B input terminals on the module.
- For the battery-integrated build: remove the original battery spring terminals (neutralise any alkaline corrosion with white vinegar then IPA; inspect underlying pads and traces). Mount the power daughter board behind the battery bay. Balance the three LiPo cells to within ~30–50mV of resting voltage, then plug them into the daughter board. Wire the daughter board's regulated 6V output to the original battery-positive node (battery side of S901) and GND to chassis. Connect the board-to-board link (B+1, GND, VBAT_SENSE, CHG/PGOOD, optional USB D+/D-). Route the daughter board USB-C to a panel-accessible location. Leave the ~4.6mm cell swell clearance unobstructed.
- Reassemble partially (leave accessible for power-on test). Connect power. Insert test tape. Press play. Verify correct speed; adjust RV601 if necessary.
- Close the case. Installation complete.

10. Bill of Materials

10.1 Shared Core Components

Ref	Part	Description	Qty
U1	STM32G0C1KCU6	MCU — UFQFPN32, 64MHz, 256KB flash (GP variant)	1
U2	MT3608	Boost converter SOT-23-6 — B+1 6V → B+3 10.8V	1
U3	MCP1700T-3302E/TT	LDO SOT-23-3 — B+1 6V → 3.3V MCU supply	1
U4	TPS63070RNLT	Buck-boost SOT-23-8 — V_RAIL → B+1 6.0V (wall variants)	1
D1	USBLC6-2SC6Y	USB ESD protection SOT-23-6 — D+/D- clamp	1
J1	8-pos 1.25mm JST	Machine interface harness connector	1
L1	4.7μH inductor	MT3608 boost inductor, 4×4mm footprint, 1.5A	1
L2	TPS63070 inductor	Per TPS63070 datasheet; ~2.2μH typical	1
R1/R2	649kΩ / 100kΩ 0402	TPS63070 output divider — sets B+1 = 6.0V	1 each

10.2 Variant A Components (DNP for Variant B)

Ref	Part	Description	Qty
U5	IP2721	USB PD trigger SOT-23-6 — requests 9V PDO	1
J2	USB-C receptacle	Panel-mount or flex-tail USB-C connector	1

10.3 Variant B Components (DNP for Variant A)

Ref	Part	Description	Qty
U6	LTC4359IS8	Ideal diode bridge controller SO-8	1
Q1–Q4	P-channel MOSFETs	LTC4359 bridge MOSFETs — 4× PFET, dual SO-8	2 (dual pkg)
F1	500mA polyfuse	MF-R050 or equiv, resettable	1
D3	SMBJ7.0A	TVS diode SMA — 7V standoff bidirectional	1
C_bulk	220μF 10V electrolytic	Input bulk storage — motor surge / bridge holdover	1
J3	5.5mm barrel jack	CN301 barrel jack (retained in original panel)	1

10.4 Battery-Integrated Build — Daughter Board Components

Ref	Part	Description	Qty
U_CHG	BQ24074RGTT	Power-path charger with DPPM — VQFN-16	1
U_PROT	DW01A SOT-23-6	LiPo pack protection IC	1

Q_FET	FS8205 SO-8	Dual N-ch FET — DW01 switch (or equiv FS8205A)	1
U_BOOST	TPS63070RNLT	Buck-boost — SYS rail → B+1 6.0V (daughter board)	1
CELLS	803040 LiPo × 3	3.7V 1000mAh each — 3P pack; on-cell PCM + PH2.0 pigtail	3
D_INJ	1N5819 SMA	Schottky — reverse protection at 6V injection point	1
J_CELLS	PH2.0 3-way header	Cell pigtail connectors (service disconnect)	3
R_ISET	1.1kΩ 0402	BQ24074 charge current — ~0.8A	1
R_ILIM	800Ω 0402	BQ24074 input current limit — ~2A	1

11. Repository Structure and Licensing

11.1 Repository Layout

wmd6c-servo/	
├─ hardware/	KiCad schematic and PCB files
├─ firmware/	STM32 bare-metal C source
├─ docs/	This datasheet and supporting documents
├─ LICENSE	CERN OHL-P v2 hardware licence
├─ LICENSE-FIRMWARE	MIT licence
└─ README.md	Quick-start guide

11.2 Licensing

Hardware design files (schematics, PCB layout, BOM) are licensed under the CERN Open Hardware Licence Permissive v2 (CERN OHL-P v2), which permits use, modification, and distribution with attribution and source availability retained.

Firmware source code is licensed under the MIT License, which permits free use, modification, and distribution without restriction.

Documentation (this datasheet, block diagram, application notes) is licensed under Creative Commons CC BY 4.0, requiring attribution.

11.3 Contributing

Contributions are welcome via pull request at the repository. Please include test data from at least one physical WM-D6C unit for hardware changes. Firmware contributions must maintain the bare-metal, no-HAL, no-RTOS, no-malloc constraints of the existing codebase.

12. Electrical Characteristics and Performance

12.1 Absolute Maximum Ratings

Parameter	Value	Notes
Supply voltage on B+1 rail (wall variants)	6.5V max	TPS63070 operating range ceiling
Input voltage Variant A (before IP2721)	20V max	PD source; do not exceed

Input voltage Variant B (before bridge)	9V operating; abs max 14V	TVS begins conducting above ~7V standoff
STM32 VDD supply voltage	3.6V max	Below this all I/O are safe
ADC / GPIO input voltage (3.3V logic)	VDD + 0.3V = 3.6V max	BAT54 clamps on FG_IN provide protection
Motor back-EMF transient (B+3 rail)	Clamped to ~12V by SMBJ7.0A	Brief transient only
Operating temperature	0°C to +50°C	Limited by LiPo cell specification in battery build
Storage temperature	-20°C to +60°C	Consistent with LiPo cell storage limits

12.2 Operating Characteristics

Parameter	Typical	Notes
B+1 output voltage (all variants)	6.0V ± 0.15V	TPS63070 regulation; 0.8V reference ±1%
B+3 output voltage	10.8V ± 0.3V	MT3608; R_upper/R_lower divider
VDD output voltage	3.3V ± 0.03V	MCP1700 regulation
Total B+1 current (machine at full load)	~550–650mA	Audio circuits + servo logic + Auto-Off board
B+3 current (motor at operating speed)	~250–350mA steady state	Peaks to ~500mA during acceleration
Module quiescent current (standby, USB unplugged)	< 5mA	Dominated by TPS63070 quiescent (~55µA) and MCP1700 (~1.6µA)
TPS63070 peak efficiency (6V out, V_RAIL ≈ 4.5V)	~93%	From TI datasheet at this operating point
MT3608 efficiency (6V in, 10.8V out, 300mA)	~85%	Typical for non-synchronous boost at this point
BQ24074 charge dissipation (5V in, 4.0V bat, 0.8A)	~0.8W	Linear charger; use copper pour under IC

12.3 Expected Speed Performance

Parameter	Expected Value
Tape speed accuracy (play)	±0.05% from set point under normal operating conditions
Wow & flutter (W&F)	≤ 0.05% WRMS on a fully serviced machine; ceiling ~0.03% WRMS after optimization
Servo loop rate	~2500 Hz (TIM2 input capture ISR driven by FG pulses)
FG capture resolution	~15.6 ns per tick at 64 MHz; floor ~0.0039% WRMS — not the limiting factor
PI loop execution time	< 500 ns per cycle (fixed-point Q16 arithmetic, no float)
Digital measurement floor	~0.0039% WRMS per TIM2 tick (the mechanical transport is the ceiling)

Motor supply (B+3)	$10.8V \pm 0.3V$, < 50mV peak-to-peak ripple at rated load
MCU supply (VDD)	$3.3V \pm 0.03V$